02402 Statistics (Polytechnical Foundation)

Lecture 2: Random variables and discrete probability distributions

DTU Compute
Technical University of Denmark
2800 Lyngby – Denmark

Agenda

1. Random variables and density functions
2. Distribution functions
3. Specific (discrete) distributions I: The binomial
   - Example 1
4. Specific distributions II: The hypergeometric
   - Example 2
5. Specific distributions III: The Poisson
   - Example 3
6. Distributions in Python
7. Mean and variance of discrete distributions

Overview

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Random variables

A random variable represents a value of an outcome *before* the corresponding *experiment* is carried out.

Random variables

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- A throw of a dice.
- The number of six'es in ten dice throws.
- Fuel consumption of a car.
- Measurement of glucose level in blood sample.
- ...

Discrete and continuous random variables

- We distinguish between *discrete* and *continuous* random variables.
- Discrete:
  - Number of people in this room who wear glasses.
  - Number of planes departing from CPH within the next hour.
- Continuous:
  - Wind speed measurement.
  - Transport time to DTU.
- Today: Discrete. Next week: Continuous.

Random variable

Before the experiment is carried out, we have a random variable

$$X \text{ (or } X_1, \ldots, X_n)$$

indicated with capital letters.

Kahoot!

(x1)

Random variable

Before the experiment is carried out, we have a random variable

$$X \text{ (or } X_1, \ldots, X_n)$$

indicated with capital letters.

After the experiment is carried out, we have a *realization* or *observation*

$$x \text{ (or } x_1, \ldots, x_n)$$

indicated with lowercase letters.



(x1)

Python: Simulate rolling a die

- Go to today's Python notebook in VS Code
  - "Example 1: Simulation of a stochastic variable (rolling a dice)"



Discrete distributions

- Random variables are used to describe an experiment before it is carried out.
- How to describe the possible future without yet knowing the outcome?

Discrete distributions

- Random variables are used to describe an experiment before it is carried out.
- How to describe the possible future without yet knowing the outcome?
- Solution: Use a *probability mass function (PMF) and cumulative distribution function (CDF)*.

A ***probability mass function (PMF)*** describes the probability of each of the different outcomes of the experiment.

Classification of probability distributions

Concept	**SciPy**	**Discrete**	**Continuous**
Mass / density	`.pmf` / `.pdf`	`pmf`	`pdf`
Cumulative probability	`.cdf`	✓	✓
Quantile (inverse cdf)	`.ppf`	✓	✓
Simulate random values	`.rvs`	✓	✓

Density function, discrete random variable: Definition 2.6

The *density function* (probability density function, pdf) of a discrete random variable:

Definition

$$f(x) = P(X = x)$$

Describes the probability that X takes the value x when the experiment is carried out.

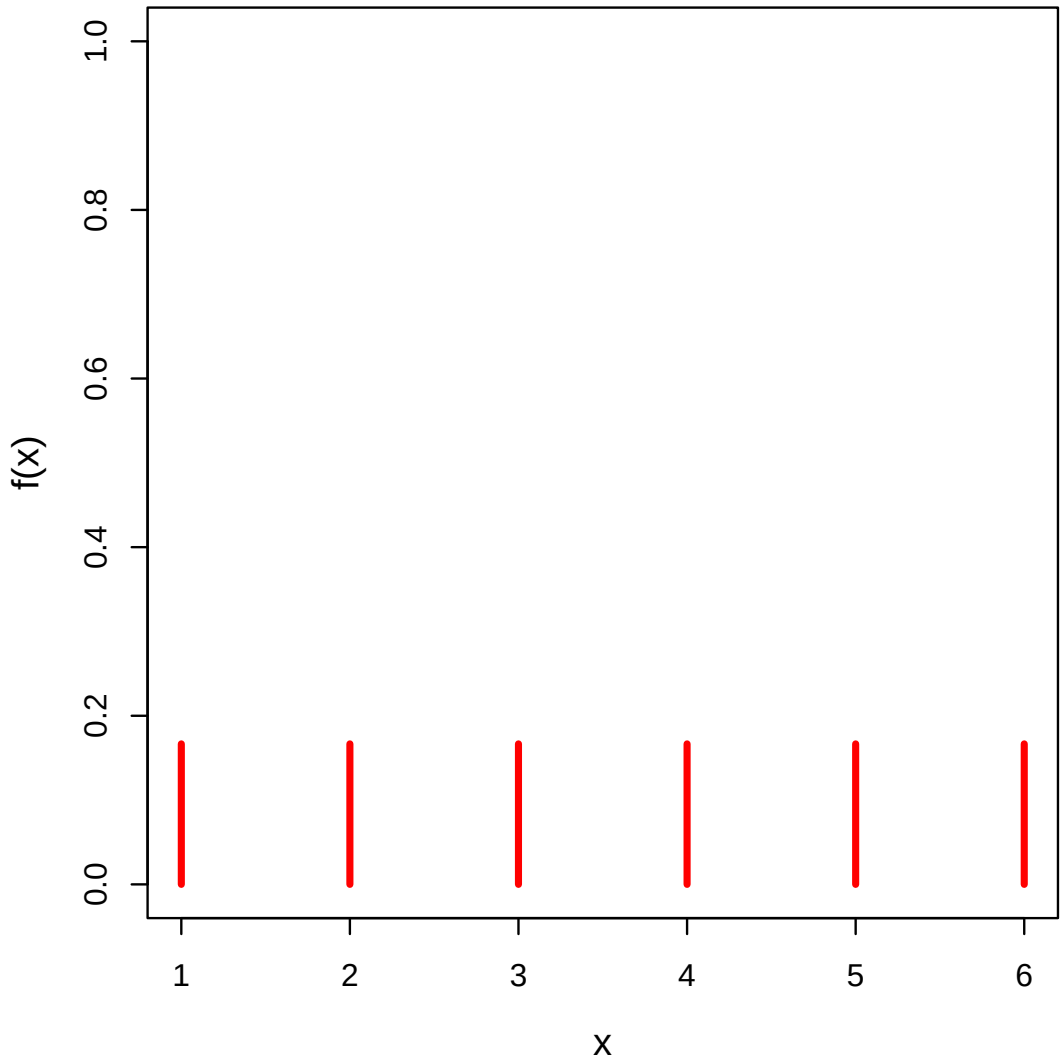
Density function, discrete random variable, Definition 2.6

The density function of a discrete random variable satisfies two properties:

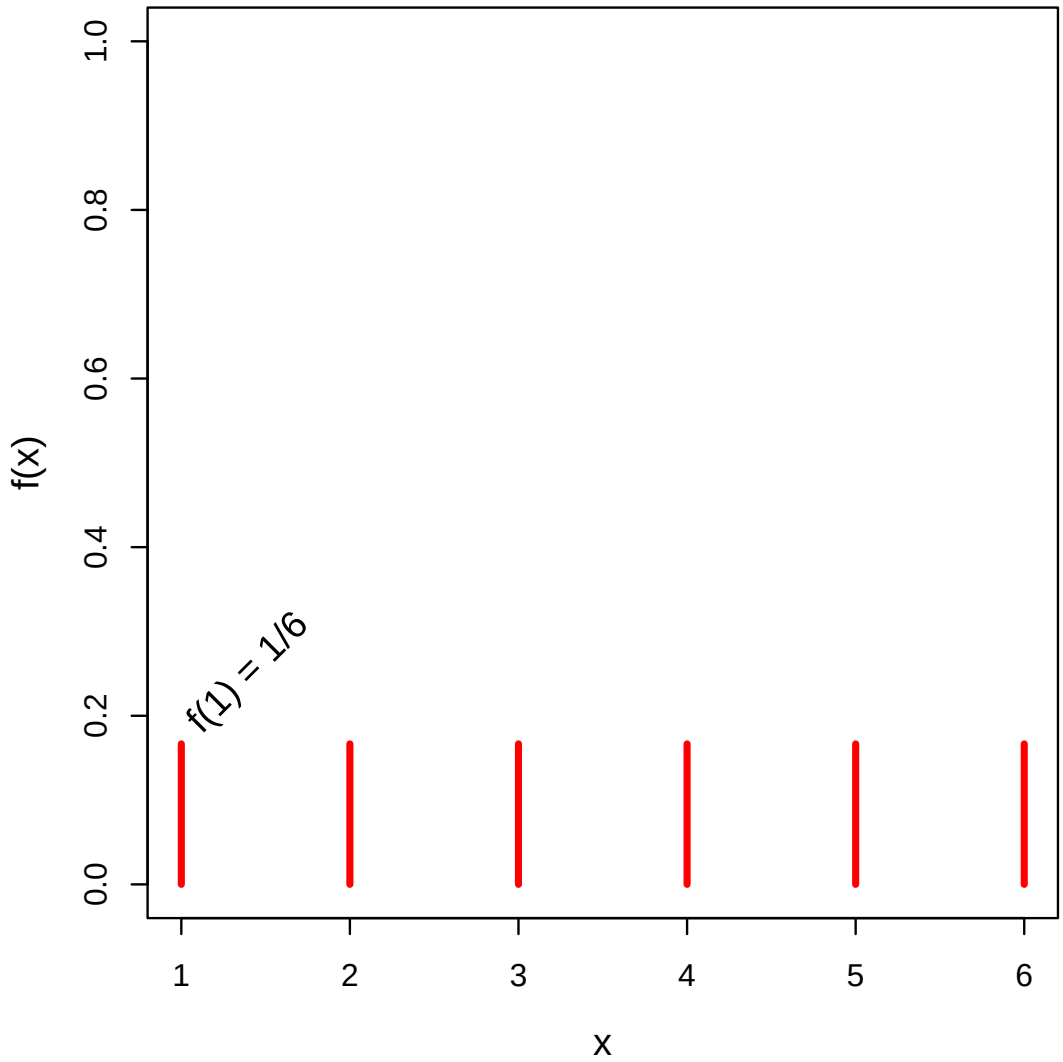
Definition

$$f(x) \geq 0 \text{ for all } x \quad \text{and} \quad \sum_{\text{all } x} f(x) = 1$$

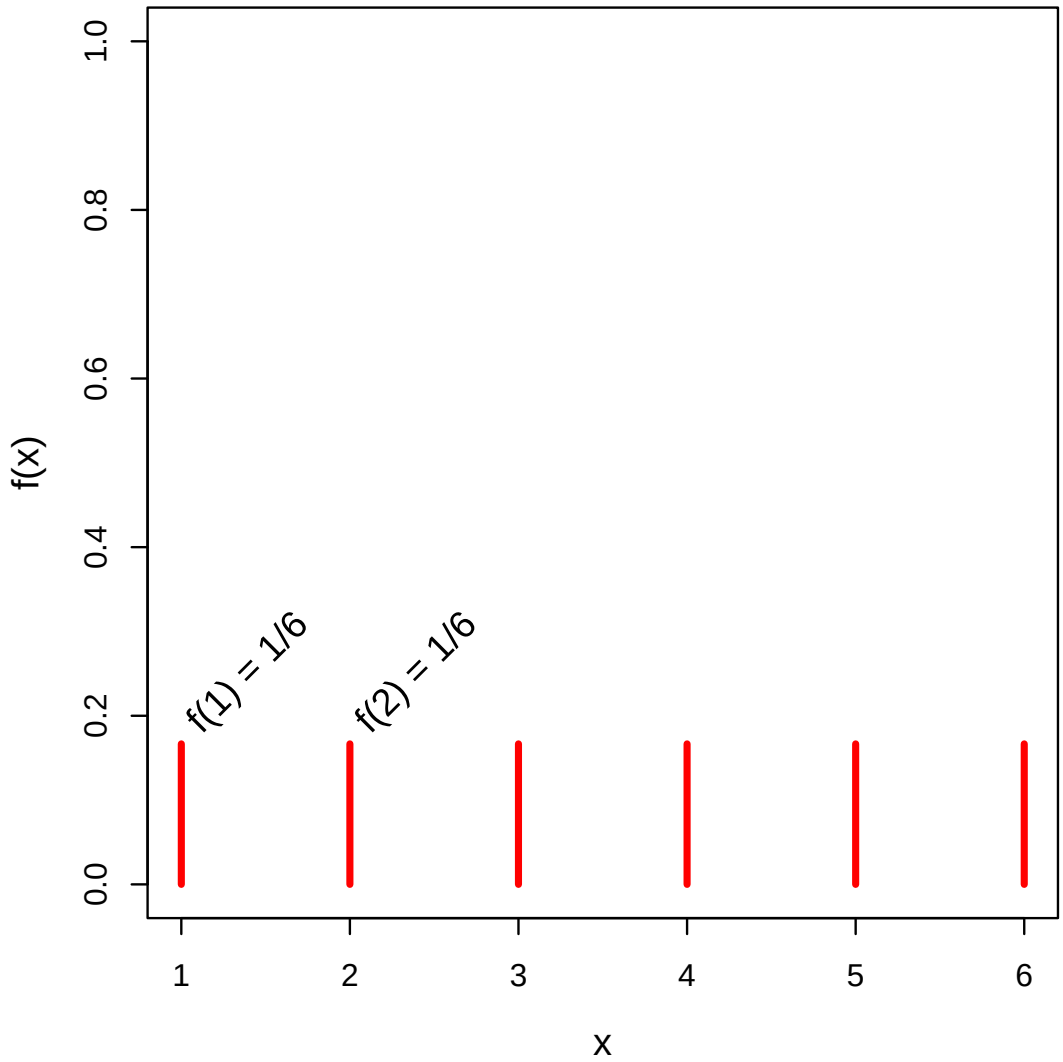
Density function for a fair dice



Density function for a fair dice



Density function for a fair dice



Sample

If we only have a single observation, can we see the distribution?

Sample

If we only have a single observation, can we see the distribution? No!

But if we have n observations, then we have a *sample*

$$\{x_1, x_2, \ldots, x_n\}$$

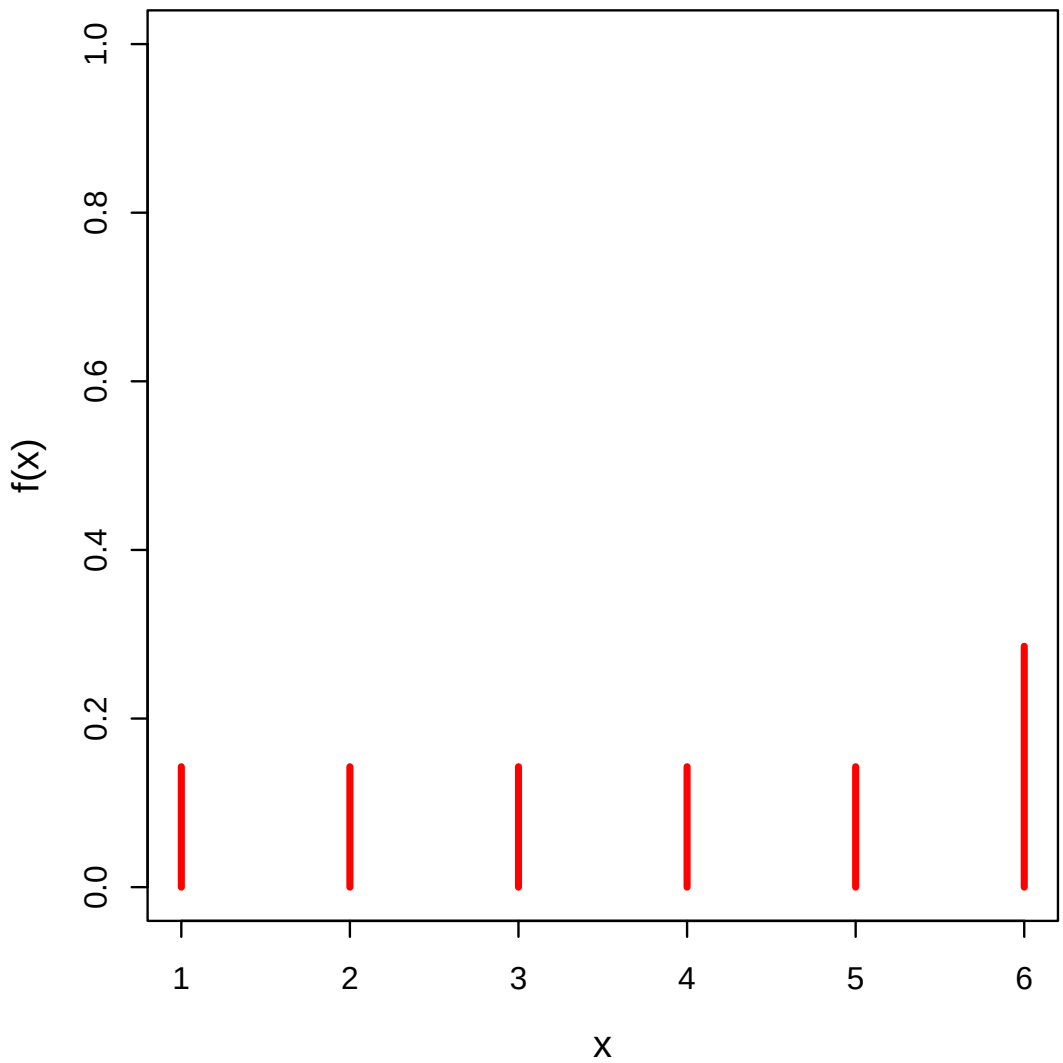
and we can begin to get an idea of the distribution.

Python: Simulate n rolls with a fair dice

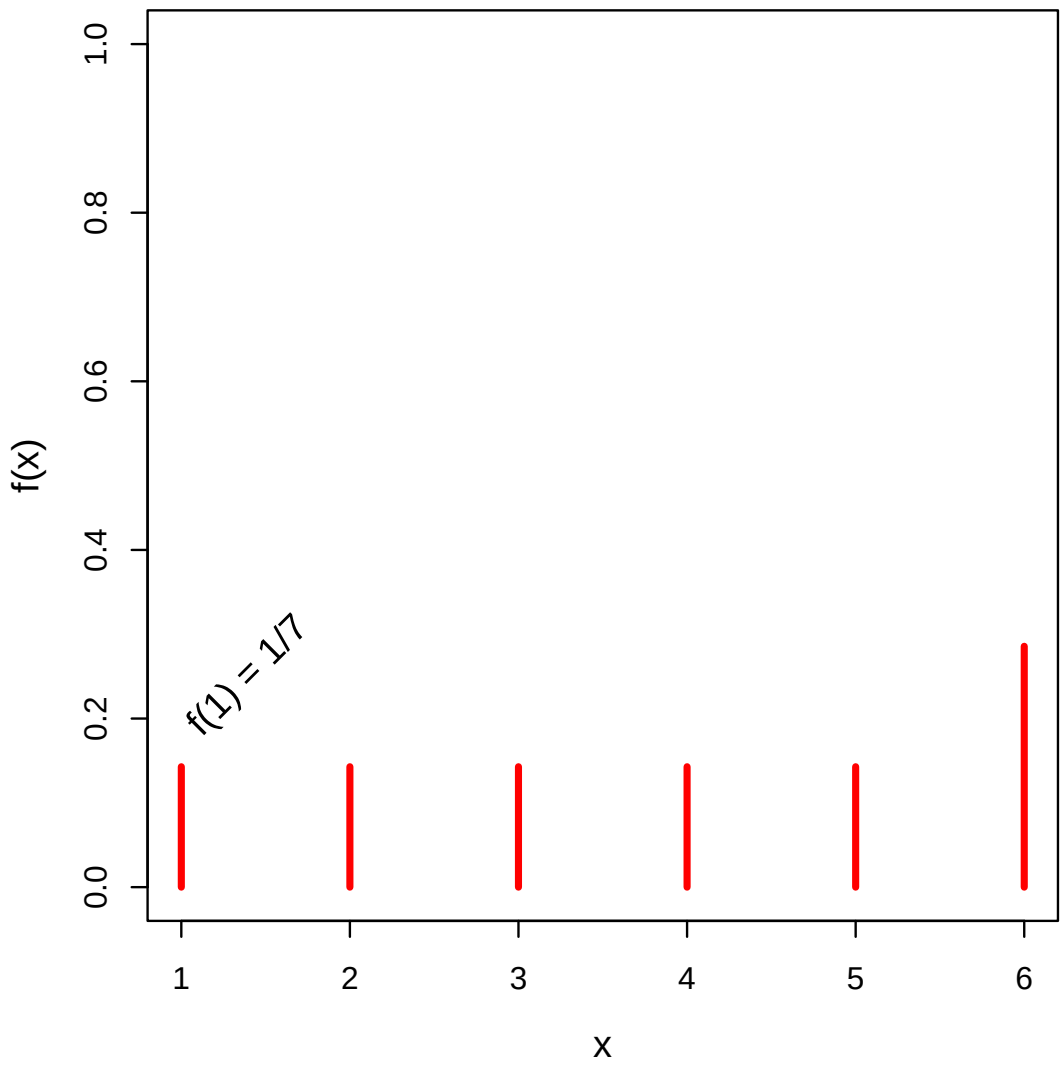
- Go to today's Python notebook in VS Code
  - "Example 2: Simulate n rolls with a fair dice"



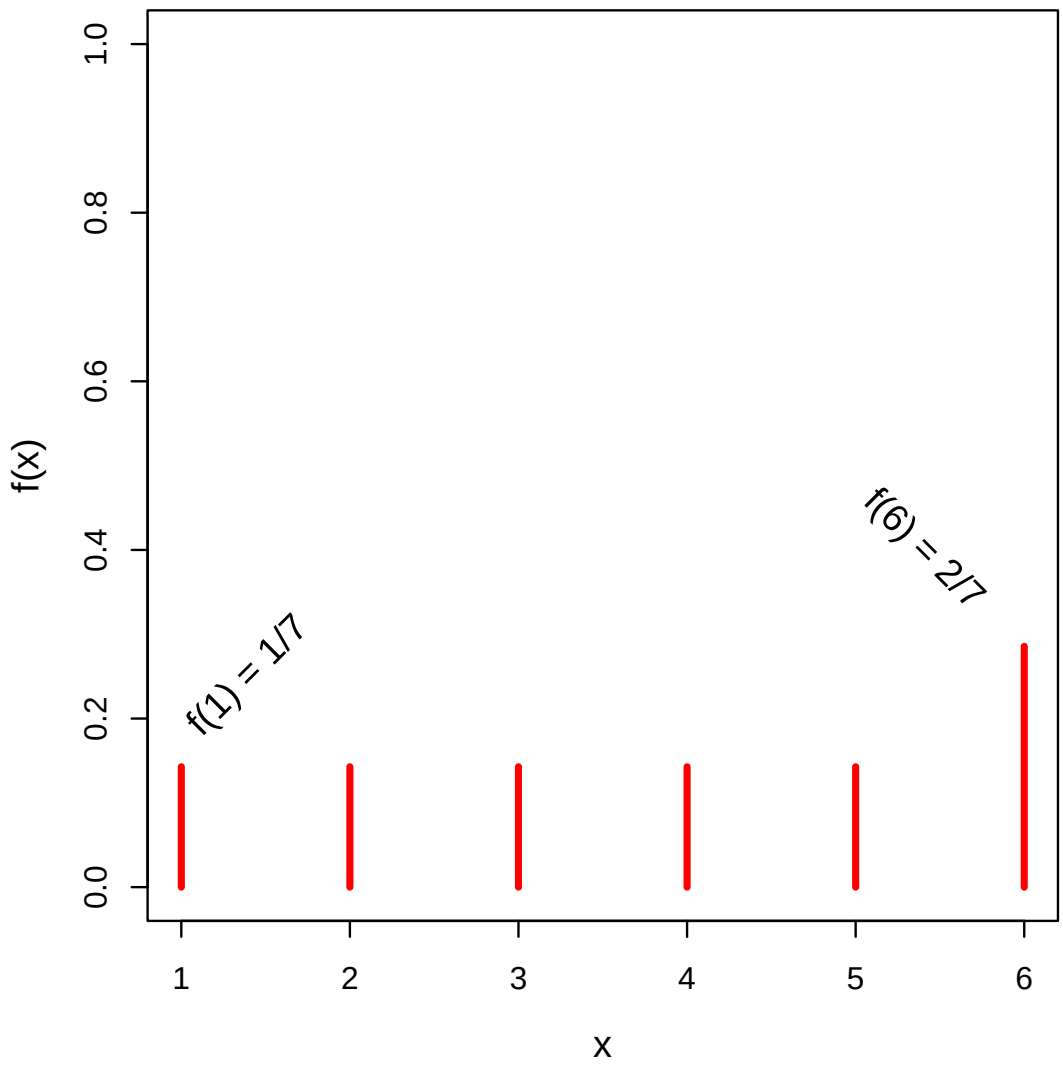
Density function for an unfair dice



Density function for an unfair dice



Density function for an unfair dice



Python: Simulate n rolls with an unfair die

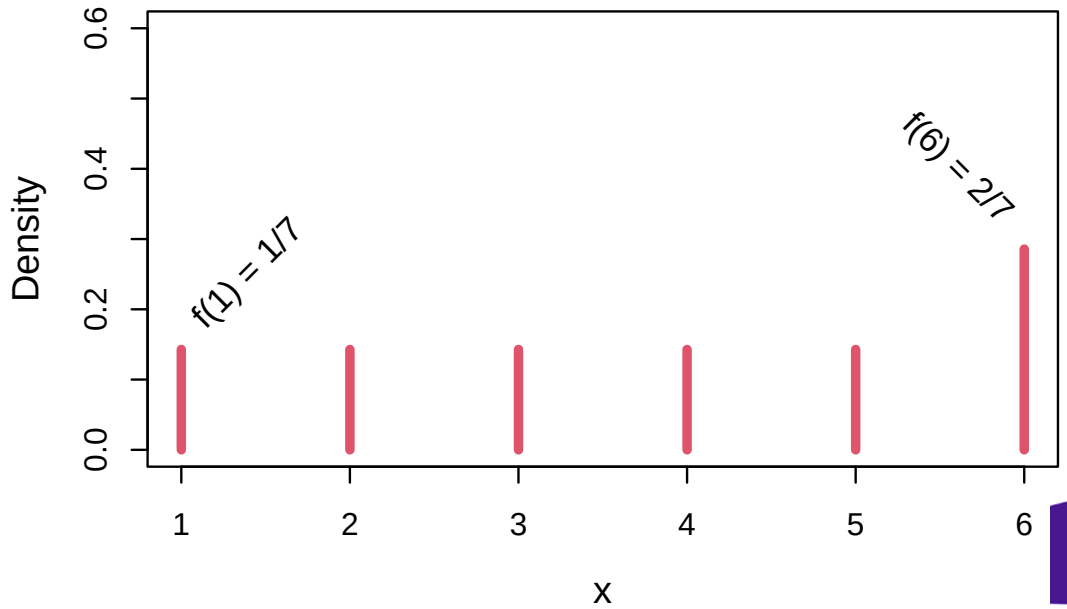
- Go to today's Python notebook in VS Code
  - "Example 3: Simulate n rolls with an **unfair** dice"



Some questions

Let X describe one throw with the *unfair* dice. What is:

- The probability of getting a 4?
- The probability of getting a 5 or a 6?
- The probability of getting less than 3?



Kahoot!

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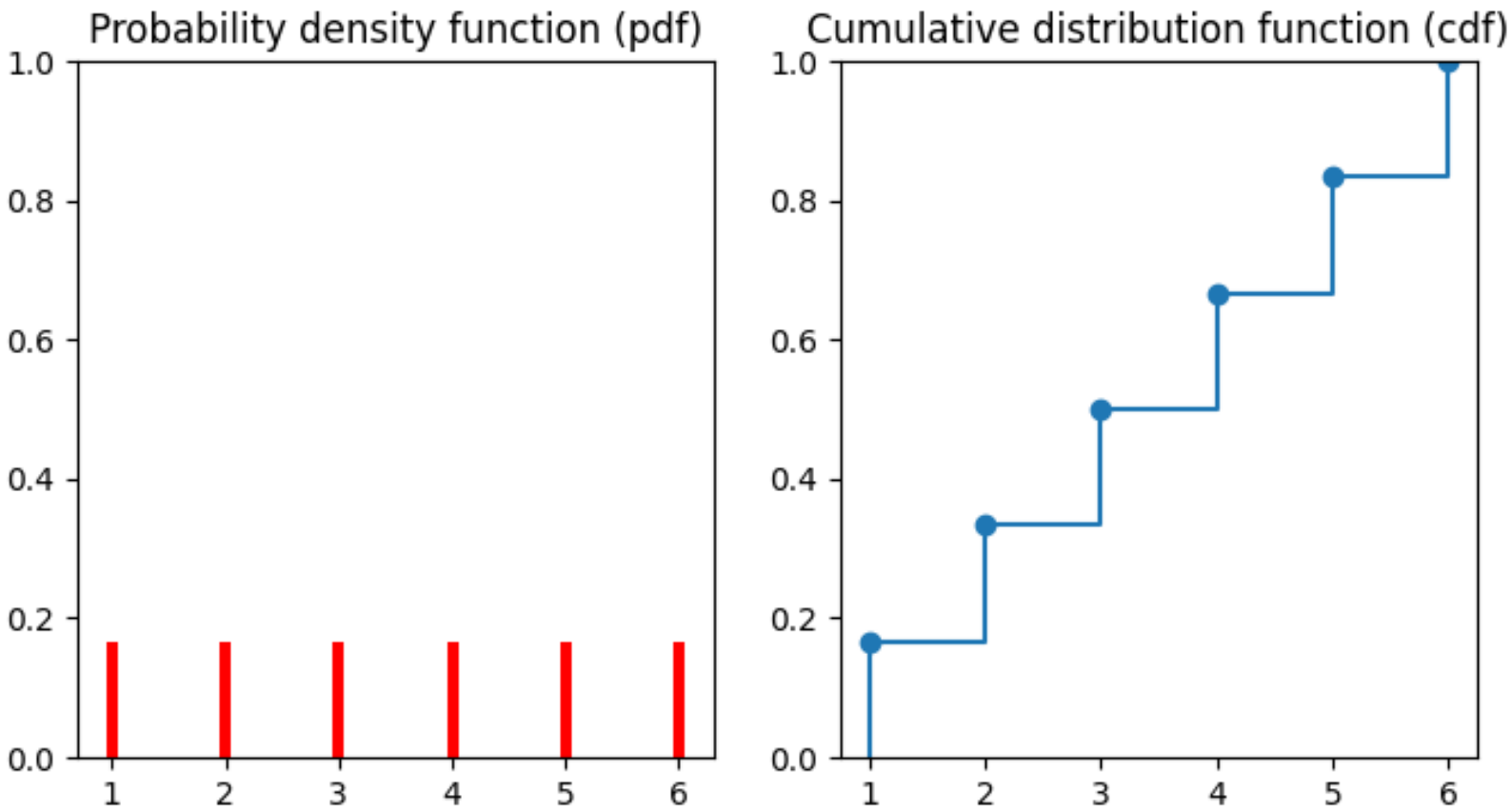
Distribution function, discrete random variable: Definition 2.9

The *distribution function* (cumulative distribution function, cdf) of a discrete random variable:

Definition

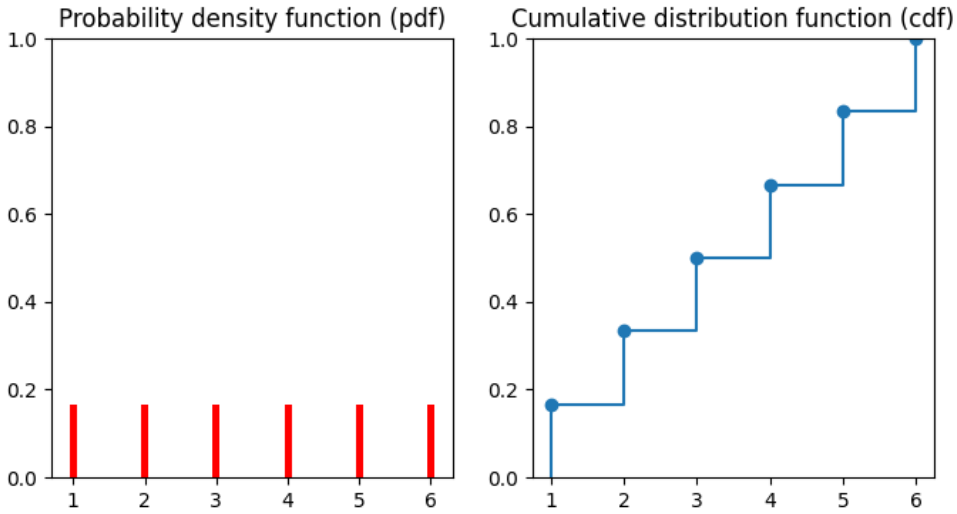
$$F(x) = P(X \leq x) = \sum_{j \text{ where } x_j \leq x} f(x_j)$$

Example: Rolling a Fair Die



Example: Roll of a fair die

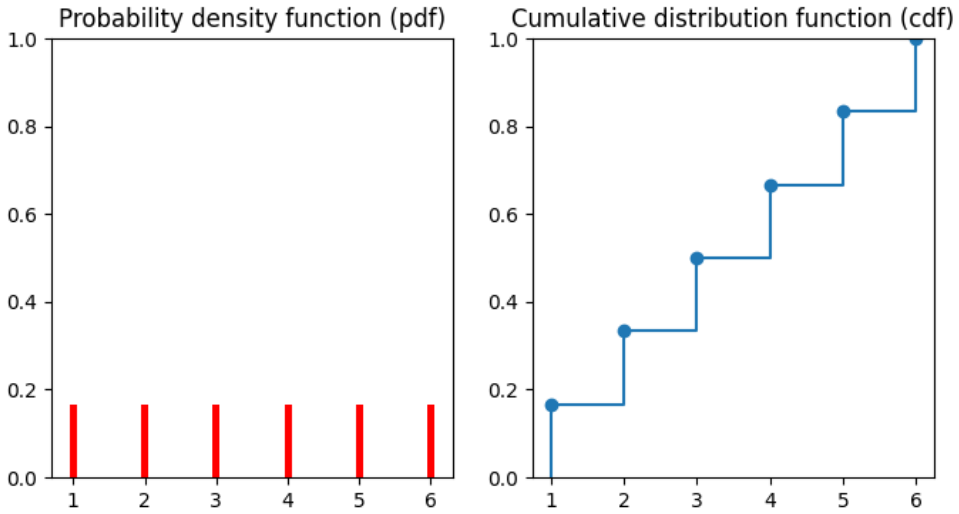
Let X represent the value of a roll with a **fair** die.



Find the probability of observing a value less than 3:

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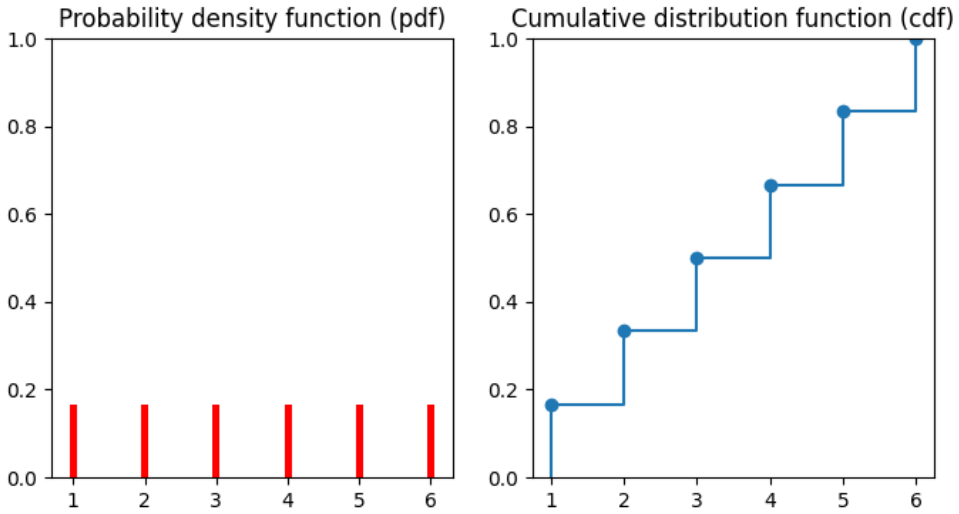


Find the probability of observing a value less than 3:

$$P(X < 3)$$

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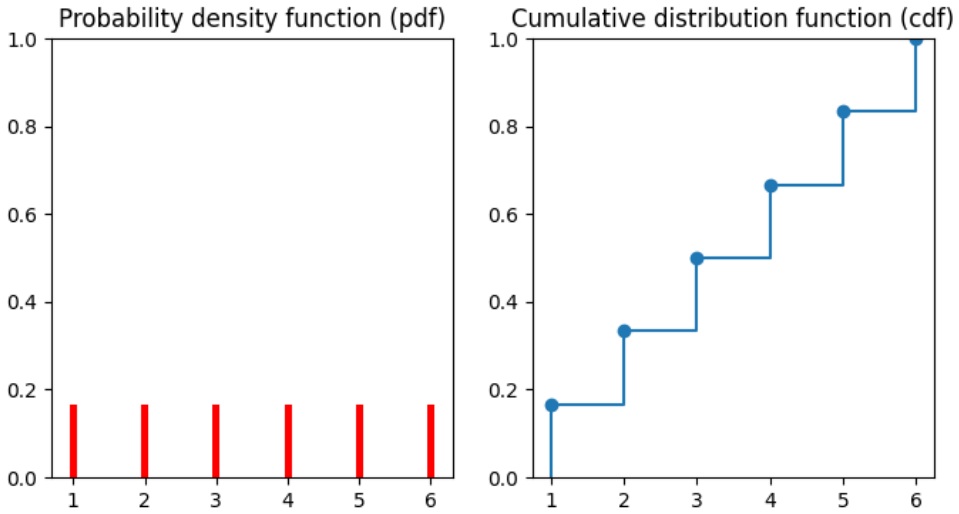


Find the probability of observing a value less than 3:

$$P(X < 3) = P(X \leq 2)$$

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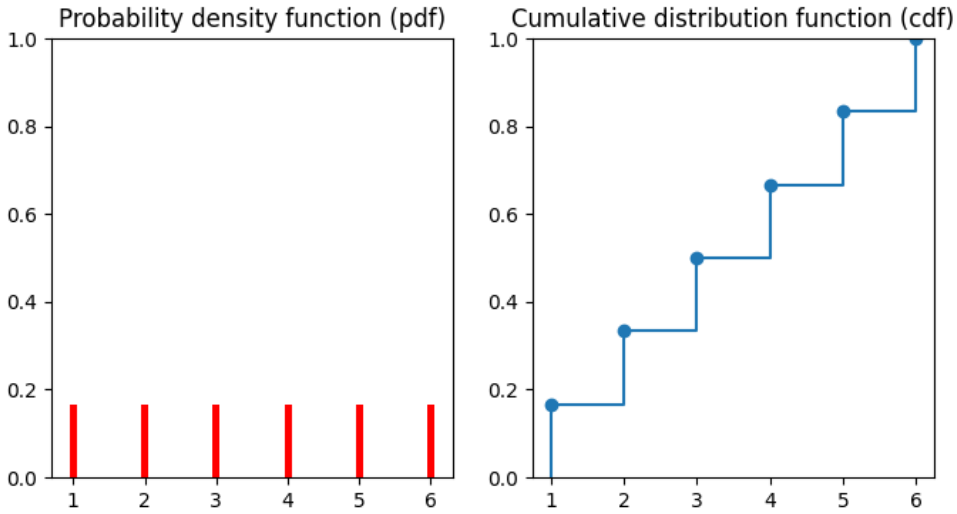


Find the probability of observing a value less than 3:

$$P(X < 3) = P(X \leq 2)$$
$$= F(2) \text{ } \textit{cumulative distribution function}$$

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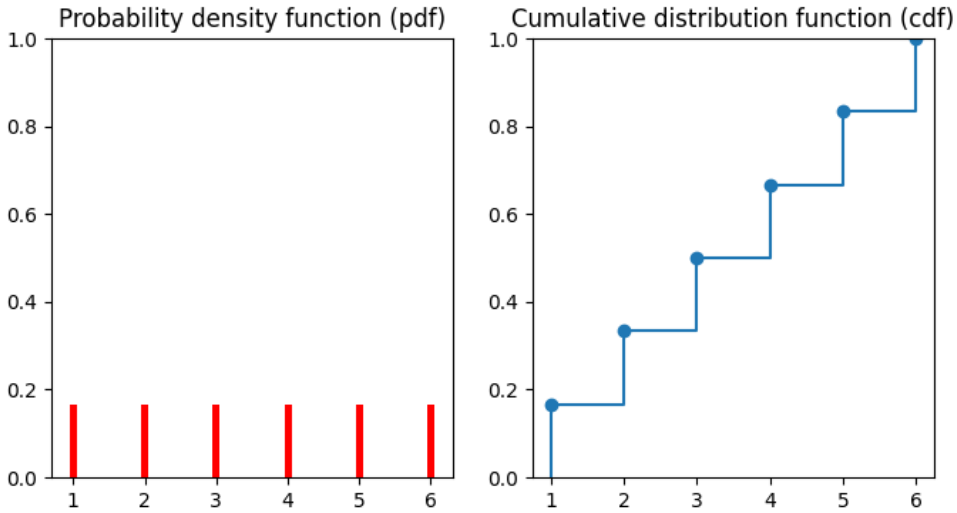


Find the probability of observing a value less than 3:

$$\begin{aligned} P(X < 3) &= P(X \leq 2) \\ &= F(2) \quad \textit{cumulative distribution function} \\ &= P(X = 1) + P(X = 2) \end{aligned}$$

Example: Roll of a fair die

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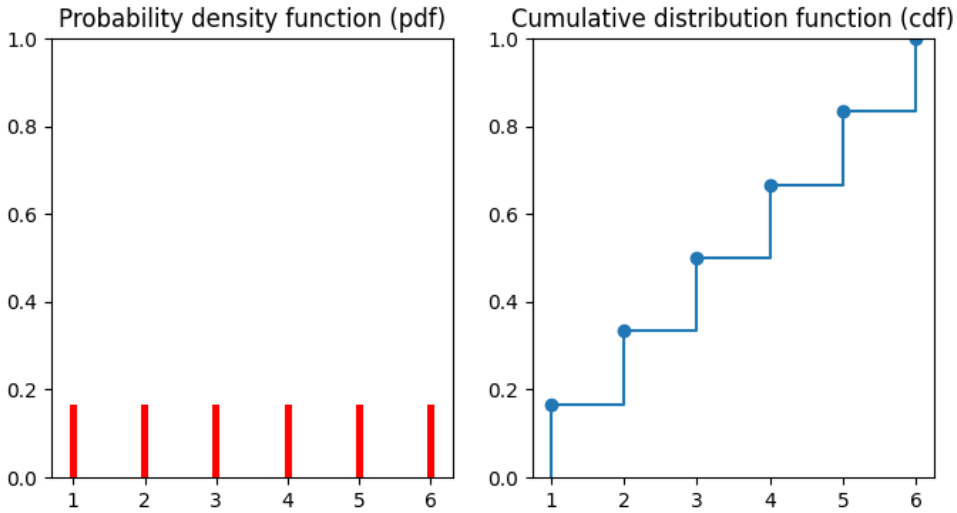


Find the probability of observing a value less than 3:

$$
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P(X < 3) &= P(X \leq 2) \\
&= F(2) \quad \textit{cumulative distribution function} \\
&= P(X = 1) + P(X = 2) \\
&= f(1) + f(2) \quad \textit{probability density function}
\end{aligned}
$$

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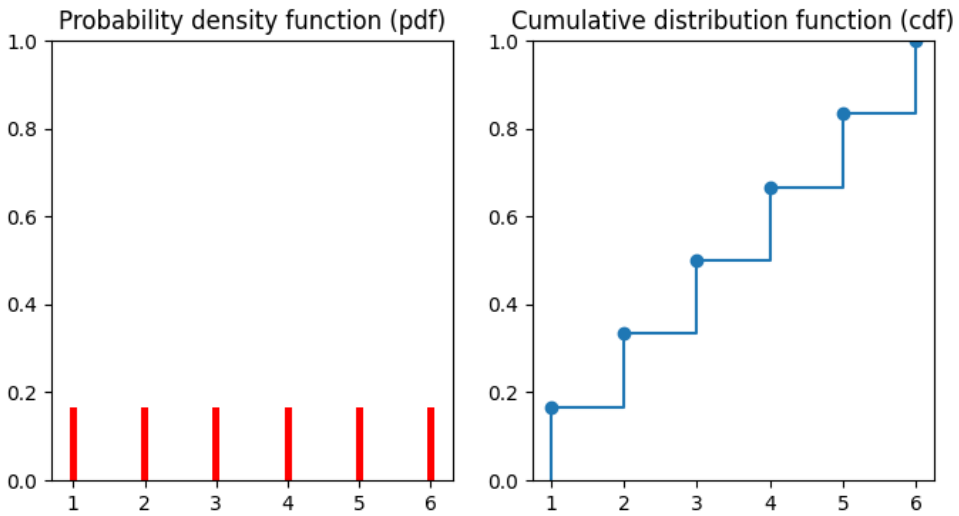


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&= P(X = 1) + P(X = 2) \\
&= f(1) + f(2) \quad \textit{probability density function} \\
&= \frac{1}{6} + \frac{1}{6} = \frac{1}{3}
\end{aligned}
$$

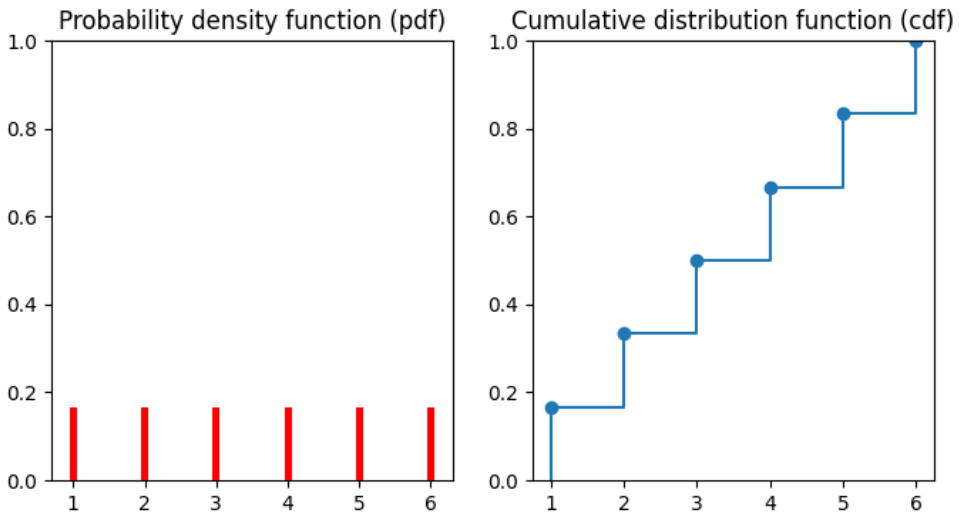
Example: Roll with a fair die

Find the probability of observing a value greater than or equal to 3:



Example: Roll with a fair die

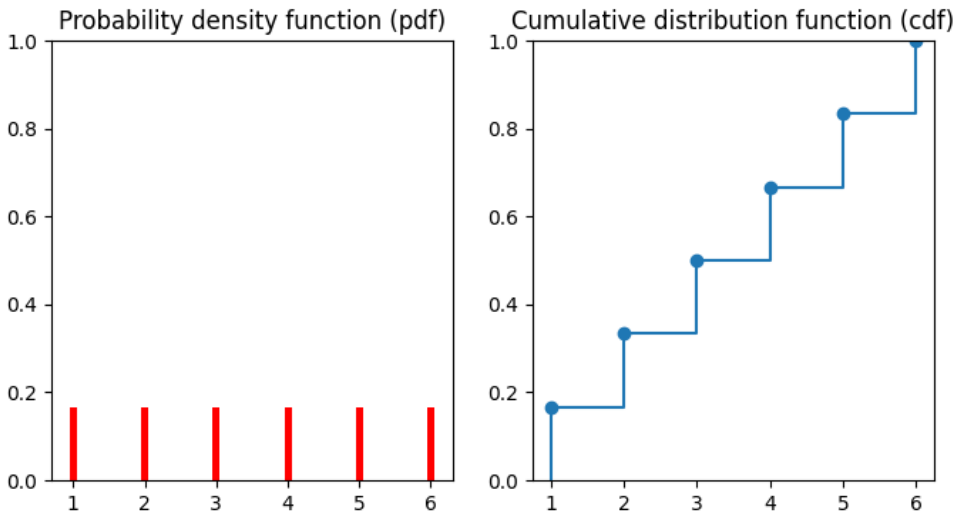
Find the probability of observing a value greater than or equal to 3:



$$P(X \geq 3)$$

Example: Roll with a fair die

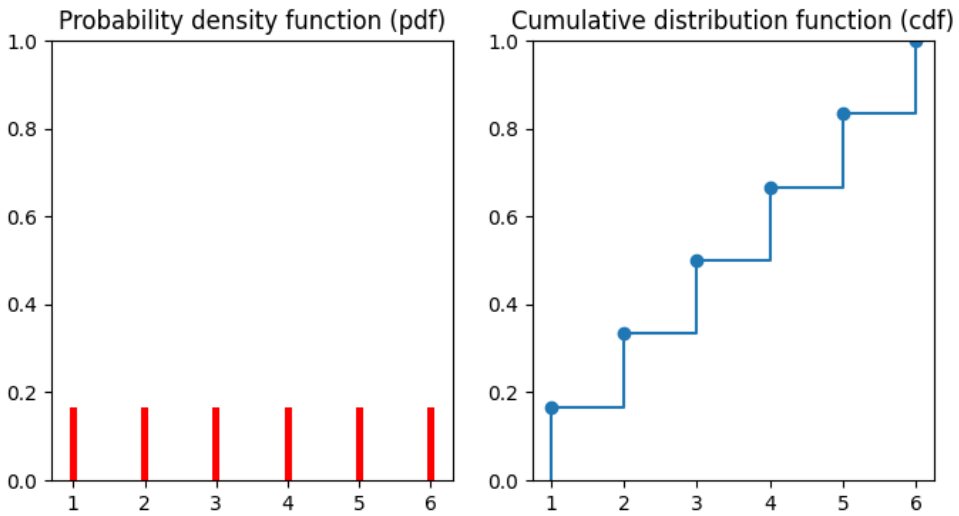
Find the probability of observing a value greater than or equal to 3:



$$P(X \geq 3) = 1 - P(X \leq 2)$$

Example: Roll with a fair die

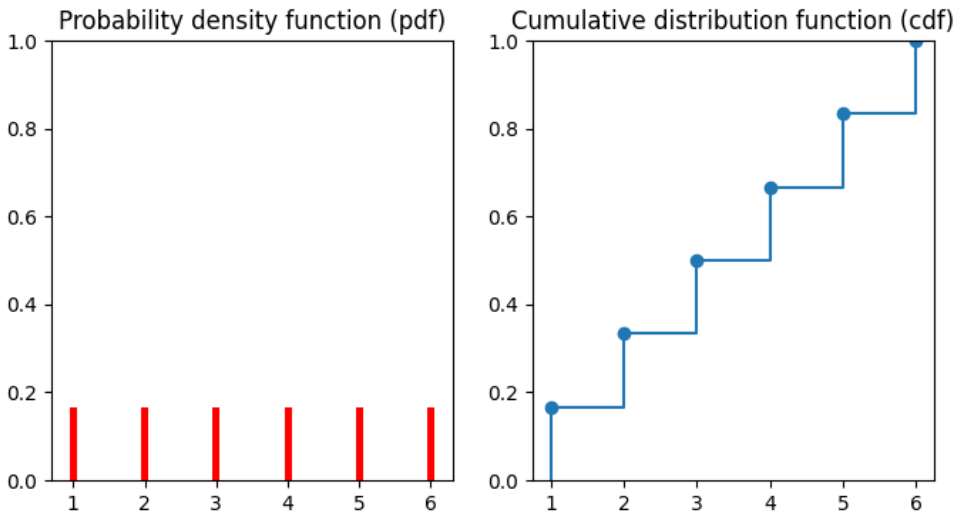
Find the probability of observing a value greater than or equal to 3:



$$P(X \geq 3) = 1 - P(X \leq 2)$$
$$= 1 - F(2) \text{ } \textit{cumulative distribution function}$$

Example: Roll with a fair die

Find the probability of observing a value greater than or equal to 3:



$$
\begin{aligned}
P(X \geq 3) &= 1 - P(X \leq 2) \\
&= 1 - F(2) \quad \textit{cumulative distribution function} \\
&= 1 - \frac{1}{3} = \frac{2}{3}
\end{aligned}
$$

Python: cdf and ecdf

- Go to today's Python notebook in VS Code
  - "Example 4: ECDF"



Kahoot!

(x2)

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Specific discrete distributions

- A number of different statistical distributions exist, which may be used to describe and analyse different types of problems.
- Today, we consider only discrete distributions:
  - The binomial distribution
  - The hypergeometric distribution
  - The Poisson distribution

The Binomial distribution

- An experiment with two outcomes, "success" or "failure", is repeated (independent repetitions).
- X is the number of successes after n repetitions.

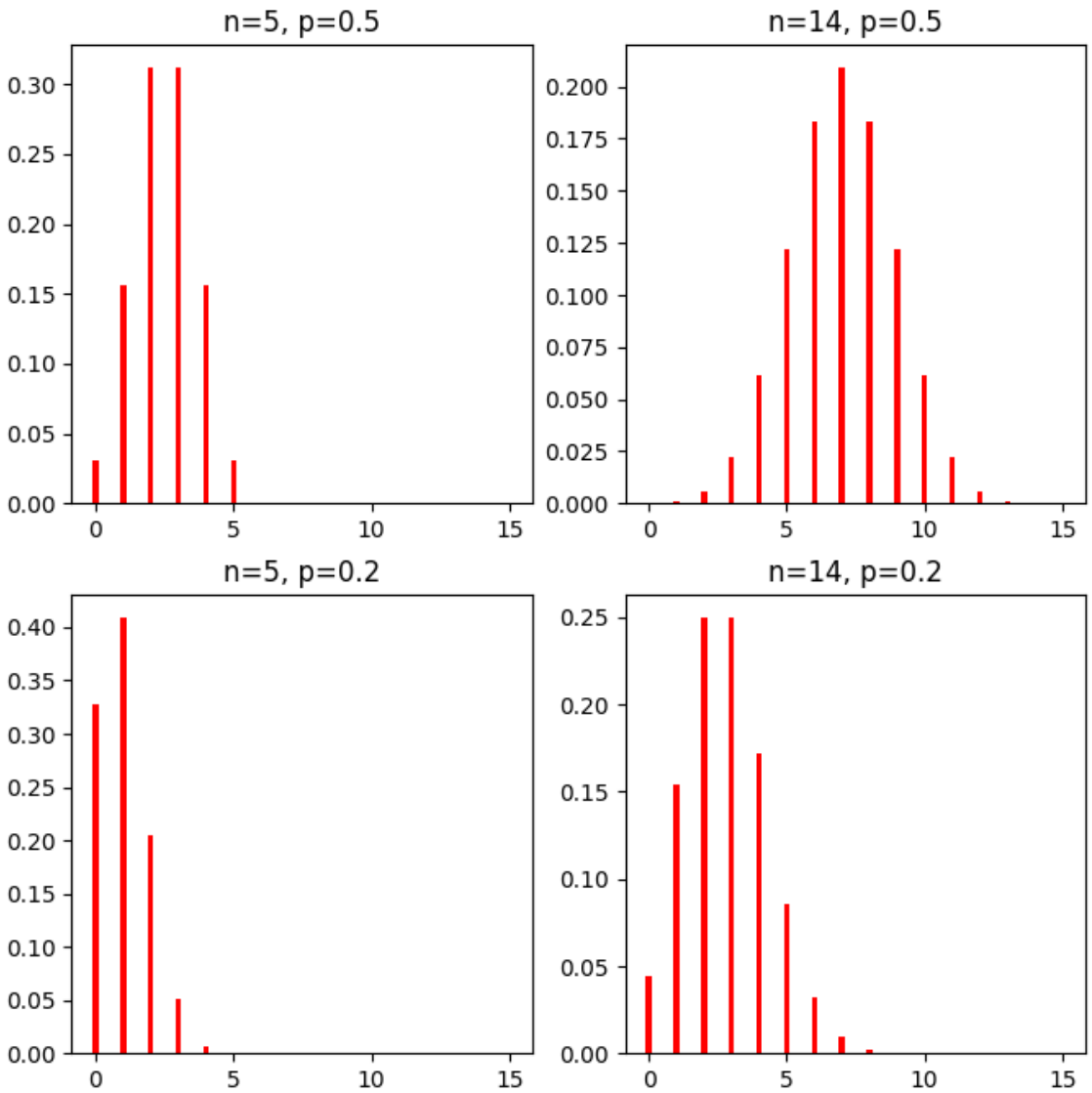
The Binomial distribution

- An experiment with two outcomes, "success" or "failure", is repeated (independent repetitions).
- X is the number of successes after n repetitions.
- Then X follows a binomial distribution:

$$X \sim B(n, p)$$

  - n: number of repetitions
  - p: probability of success in each repetition

Examples of Binomial Distributions



Binomial Distribution: Probability Mass Function

Definition 2.20 Binomial distribution

Let the random variable X be binomial distributed

$$X \sim B(n, p), \tag{2-19}$$

where n is number of independent draws and p is the probability of a success in each draw.
The binomial *pdf* describes probability of obtaining x successes

$$f(x; n, p) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \tag{2-20}$$

where

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}, \tag{2-21}$$

is the number of distinct sets of x elements which can be chosen from a set of n elements. Remember that $n! = n \cdot (n-1) \cdot \ldots \cdot 2 \cdot 1$.

Example: Binomial distribution

Suppose that $X \sim B(4, p)$, i.e. $n = 4$. Find the probability of 3 successes.

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- Probability of 3 successes: $P(X = 3)$.
- Three successes can be obtained in four "ways":
  SSSF, SSFS, SFSS, FSSS.

Example: Binomial distribution

Suppose that $X \sim B(4, p)$, i.e. $n = 4$. Find the probability of 3 successes.

- Probability of 3 successes: $P(X = 3)$.
- Three successes can be obtained in four "ways":
  SSSF, SSFS, SFSS, FSSS.
- Thus,

$$\binom{n}{x} = \binom{4}{3} = \frac{4!}{3!(4-3)!} = \frac{4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1 \cdot 1} = 4,$$

  and

$$P(X = 3) = 4p^3(1-p).$$

Python: Simulations with the Binomial Distribution

- Go to today's Python notebook in VS Code
  - "Example 5: The binomial distribution"
  - "Example 6: Number of 6's with fair dice"



Example 1

In the call center of a phone company, customer satisfaction is an issue. It is especially important that when errors/faults occur, they are corrected within the same day.

Assume that six errors occur, and that the probability of any error being corrected within the same day is 70%. **What is the probability that all six errors are corrected within the same day that they occurred?**

Example 1

In the call center of a phone company, customer satisfaction is an issue. It is especially important that when errors/faults occur, they are corrected within the same day.

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- Step 1) What should be described by a random variable X?

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- Step 2) What is the distribution of X?

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- Step 1) What should be described by a random variable X?

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- Step 2) What is the distribution of X?

  A binomial distribution with $n = 6$ and $p = 0.7$.

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  $P(X = 6) = f(6; 6, 0.7)$



Visual Studio Code

Kahoot!

(x2)

Kahoot Explained: Exactly One Defective Machine — Binomial View

Let

$$a = \text{defective} \quad (P(a) = 0.03), \qquad b = \text{non-defective} \quad (P(b) = 0.97).$$

Selecting two machines corresponds to the expansion:

$$(a+b)^2 = a^2 + ab + ba + b^2.$$

Interpretation of the terms:

- a^2: both machines are defective
- b^2: both machines are non-defective
- ab: first defective, second non-defective
- ba: first non-defective, second defective

Exactly one defective machine corresponds to the two mixed terms, probaility that exactly one or two machines is defective:

$$ab + ba = 2 \cdot (0.03)(0.97) = 0.0582$$

Number of ways: $\boxed{2}$

Guessing a multiple choice exam – Binomial

Exam question, June 2026 – courses 02323 & 02402

Setting

A student answers all 30 questions of a multiple choice exam completely at random. Each question has five options, exactly one of them correct. What is the probability of getting *at least* 15 right?

Which Python command computes this probability?

1. `1 - stats.hypergeom.cdf(14, 30, 6, 15)`
2. `1 - stats.hypergeom.cdf(15, 30, 5, 15)`
3. `1 - stats.binom.cdf(14, 30, 1/5)`
4. `1 - stats.hypergeom.pmf(15, 30, 6, 14)`
5. `1 - stats.binom.pmf(15, 30, 1/5)`
6. Don't know / No answer

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3. `1 - stats.binom.cdf(14, 30, 1/5)`
4. `1 - stats.hypergeom.pmf(15, 30, 6, 14)`
5. `1 - stats.binom.pmf(15, 30, 1/5)`
6. Don't know / No answer

Answer: option 3

Guessing gives $n = 30$ independent trials with success probability $p = 1/5$, so $X \sim \text{Bin}(30, 1/5)$ – *with* replacement, hence binomial and not hypergeometric. Since `cdf` is $P(X \leq x)$,

$$P(X \geq 15) = 1 - P(X \leq 14) \approx 0.00023.$$

Option 5 uses `pmf`, giving $1 - P(X = 15)$; the off-by-one trap is writing `cdf(15,...)` instead of `cdf(14,...)`.

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The hypergeometric distribution

- Again, X is the the number of successes, but now *without* replacement when repeating.

The hypergeometric distribution

- Again, X is the the number of successes, but now *without* replacement when repeating.
- X follows the hypergeometric distribution

$$X \sim H(n, a, N)$$

  - n is the number of draws (repetitions)
  - a is the number of successes in the population
  - N is the number of elements in the (entire) population

The Hypergeometric Distribution: Probability Mass Function

Definition 2.24 Hypergeometric distribution

Let the random variable X be the number of successes in n draws without replacement. Then X follows the hypergeometric distribution

$$X \sim H(n, a, N), \tag{2-24}$$

where a is the number of successes in the N elements large population. The probability of obtaining x successes is described by the hypergeometric *pdf*

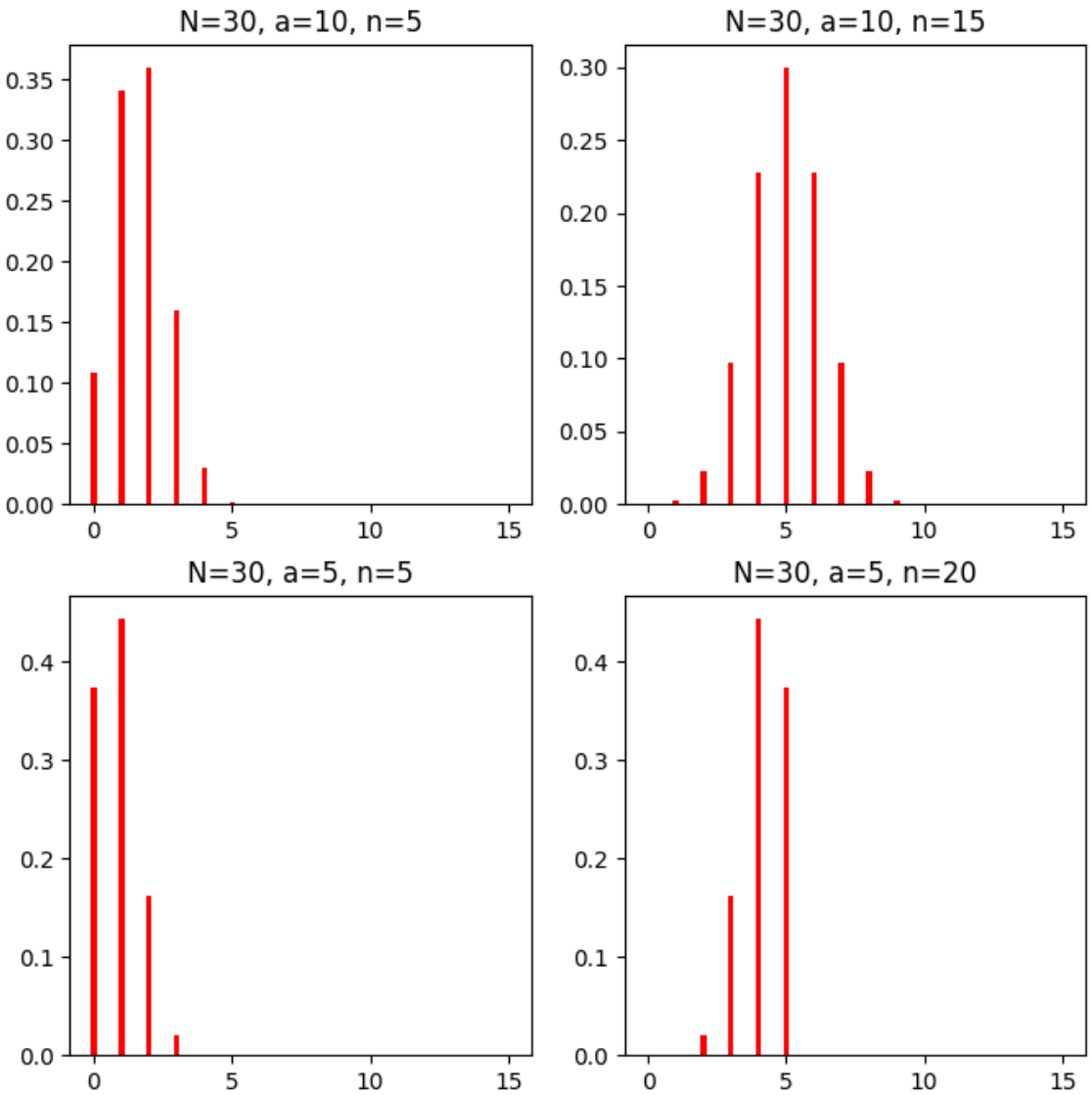
$$f(x; n, a, N) = P(X = x) = \frac{\binom{a}{x}\binom{N-a}{n-x}}{\binom{N}{n}}. \tag{2-25}$$

The notation

$$\binom{a}{b} = \frac{a!}{b!(a-b)!}, \tag{2-26}$$

represents the number of distinct sets of b elements which can be chosen from a set of a elements.

Examples on Hypergeometrical distributions



Example 2

In a shipment of 10 harddisks, 2 of them have small scratches.

A random sample of 3 harddisks is taken. **What is the probability that at least 1 of them has scratches?**

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Visual Studio Code

Binomial vs. hypergeometric

- The binomial distribution is used to analyse samples with replacement.
- The hypergeometric distribution is used to analyse samples without replacement.

Kahoot!

(x2)

The lottery jackpot – a hypergeometric count

Example exam question, May 2026 – courses 02323 & 02402

Setting

Each week the Danish state lottery draws 7 numbers at random *without replacement* from 1 to 36. A ticket carries 7 distinct numbers; the jackpot requires all 7 to match, in any order.

What is the probability of winning the jackpot with a single ticket?

1. 1 in 78,364,164,096
2. 1 in 42,072,307,200
3. 1 in 2,176,782,336
4. **1 in** 8,347,680
5. 1 in 5,040
6. Don't know / No answer

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Answer: option 4

Let X = number of matches: hypergeometric with $n = 7$, $a = 7$, $N = 36$.

$$P(X = 7) = \frac{\binom{7}{7}\binom{29}{0}}{\binom{36}{7}} = \frac{1}{\binom{36}{7}} = \frac{7!\,29!}{36!} = \frac{1}{8{,}347{,}680}.$$

Distractor 42,072,307,200 is the *ordered* count $36 \cdot 35 \cdots 30$; 5,040 is $7!$ alone.

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The Poisson distribution

- The Poisson distribution is often used as a distribution (model) for counts, which do not have a natural upper bound.
- The Poisson distribution is often characterized by its *intensity*, which is on the form "number/unit", and often denoted λ.
- A **Poisson process** is a mathematical model used to describe **random events occurring over time (or space)** where the events happen independently and at a **constant average rate**.

The Poisson distribution

Definition 2.27 Poisson distribution

Let the random variable X be Poisson distributed

$$X \sim Po(\lambda), \tag{2-29}$$

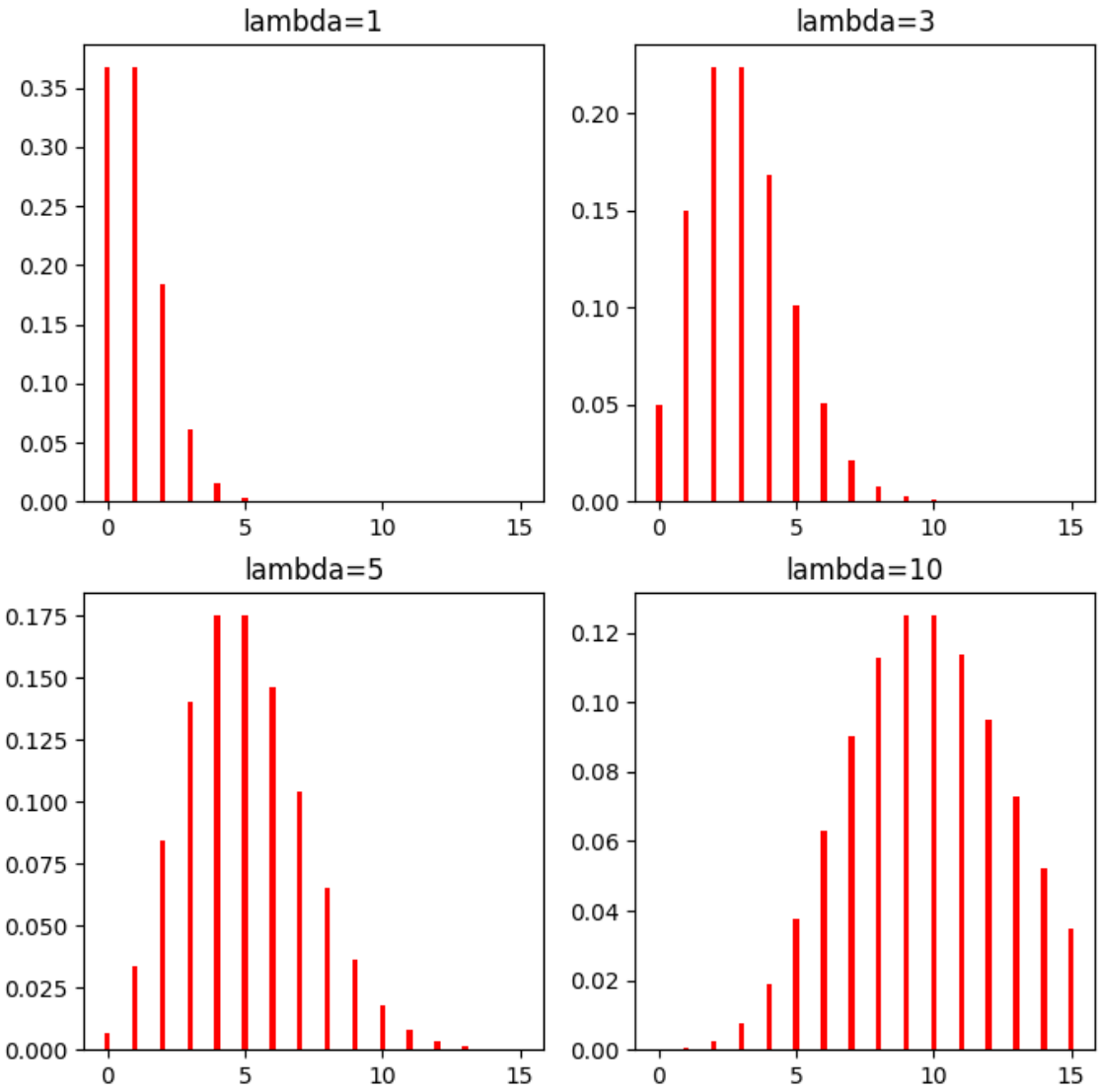
where λ is the rate (or intensity): the average number of events per interval. The Poisson *pdf* describes the probability of x events in an interval

$$f(x;\lambda) = \frac{\lambda^x}{x!}e^{-\lambda}. \tag{2-30}$$

The distribution function:

$$F(x) = P(X \leq x)$$

Examples of Poisson distributions



Example 3

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What is the probability that at most two patients are hospitalized in Copenhagen due to air pollution on any given day?

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Visual Studio Code

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Discrete Probability Distributions in Python

We use `SciPy.stats` for most distributions in this course (but also `NumPy.random.choice` for dice roll etc.).

Documentation for `SciPy.stats` can be found online (lets try it).

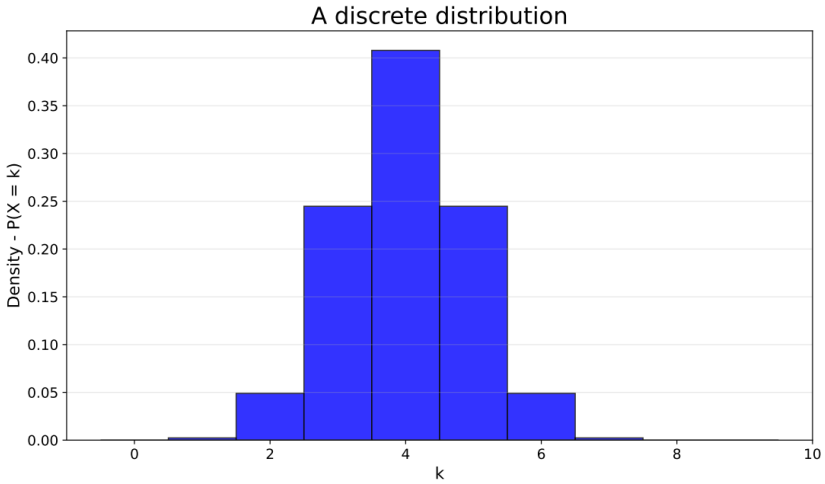
General 'methods' for most distributions are:

`scipy.stats`	`.binom/.hypergeom/.poisson`
`.rvs`	'random variates' (simulate random numbers)
`.pmf`	'probability mass function' (pmf/pdf/density function)
`.cdf`	'cumulative distribution function' (distribution function)
`.ppf`	'percent point function' (invers cdf / quantile function)
`.mean /.var /.std`	'mean'/'variance'/'standard deviation'

Example, Exam 02323 May 2026

Exercise IV

Consider the following discrete distribution on the non-negative integers, represented by its probability density function (pdf), and let X follow this distribution.



Question IV.1 (8)

Which of the following options correctly declares the distribution?

1 ☐ It is a binomial distribution with $n = 10$ and $p = 0.5$.

2 ☐ It is a binomial distribution with $n = 10$ and $p = 0.9$.

3 ☐ It is a hypergeometric distribution with $n = 7$, $a = 8$, and $N = 14$.

4 ☐ It is a Poisson distribution with $\lambda = 4$.

5 ☐ It is a Poisson distribution with $\lambda = 10$.

6 ☐ Don't know / No answer

Example, Exam 02323 May 2026

Question IV.1 (8)

Which of the following options correctly declares the distribution?

1 ☐ It is a binomial distribution with $n = 10$ and $p = 0.5$.

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5 ☐ It is a Poisson distribution with $\lambda = 10$.

6 ☐ Don't know / No answer

Since the distribution only has support on the set $\{1, ..., 7\}$ i.e., can only take values in said set, the distribution can neither be a binomial nor a Poisson distribution. In conclusion, option 3 is the correct answer.

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Mean (expectation, expected value)

Mean of a discrete random variable, Definition 2.13:

Definition

$$\mu = E(X) = \sum_{\text{all } x} x f(x)$$

- The *"true mean"* of X (as opposed to the sample mean).
- Expresses the "center" of the distribution of X.

Example: Rolling with a fair dice

Let X represent the number hit by a roll with a fair dice. Then X follows a discrete uniform distribution on the interval $[1,6]$ and has the mean:

$$\begin{aligned}\mu &= E(X) \\ &= 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} \\ &= 3.5\end{aligned}$$

Python: Compare expected value and sample mean

- Go to the Python Notebook for today in VS Code
  - "Example 7: Sample mean of n rolls with a fair dice"



Asymptotics, increasing the sample size

The more observations (the larger the sample size), the closer you get to the true mean:

$$\lim_{n \to \infty} \hat{\mu} = \mu$$

- This is called the law of large numbers.

Variance

Variance of a discrete random variable, Definition 2.16:

Definition

$$\sigma^2 = \text{Var}(X) = \sum_{\text{all } x} (x-\mu)^2 f(x)$$

- Measures average dispersion/spread.
- The "true variance" of X (as opposed to the sample variance).

Example: Rolling with a fair dice

Let X represent the number hit by a roll with a fair dice. Then X follows a discrete uniform distribution on the interval $[1,6]$ and has the variance:

$$\begin{aligned}\sigma^2 &= E[(X-\mu)^2] \\ &= (1-3.5)^2 \cdot \frac{1}{6} + (2-3.5)^2 \cdot \frac{1}{6} + (3-3.5)^2 \cdot \frac{1}{6} \\ &\quad + (4-3.5)^2 \cdot \frac{1}{6} + (5-3.5)^2 \cdot \frac{1}{6} + (6-3.5)^2 \cdot \frac{1}{6} \\ &\approx 2.92\end{aligned}$$

Python: Compare to sample variance

- Gå til dagens Python notebook i VS Code
- Go to the Python notebook for today in VS Code
  - "Example 8: Sample variance of n rolls with a fair dice"



Mean and variance of Binomial distribution

Theorem 2.21 Mean and variance

The mean of a binomial distributed random variable is

$$\mu = np, \tag{2-22}$$

and the variance is

$$\sigma^2 = np(1-p). \tag{2-23}$$

The mean and variance for the probability in the Callcenter example when calculated by hand would look like,

$$\mu = np = 6 \cdot 0.7 = 4.2$$

$$\sigma^2 = np(1-p) = 6 \cdot 0.7 \cdot (1-0.7) = 1.26$$

Mean and variance of Hypergeometric distribution

Theorem 2.25 Mean and variance

The mean of a hypergeometric distributed random variable is

$$\mu = n\frac{a}{N}, \tag{2-27}$$

and the variance is

$$\sigma^2 = n\frac{a(N-a)}{N^2}\frac{N-n}{N-1}. \tag{2-28}$$

The mean and variance for the probability in the Harddrive example when calculated by hand would look like,

$$\mu = n\frac{a}{N} = 3\cdot\frac{2}{10} = 0.6$$

$$\sigma^2 = n\frac{1(N-a)}{N^2}\frac{N-n}{N-1} = 3\frac{2(10-2)}{10^2}\frac{10-3}{10-1} = \frac{336}{900} = 0.373$$

The mean and variance of the Poisson distribution

Theorem 2.28 Mean and variance

A Poisson distributed random variable X has exactly the rate λ as the mean

$$\mu = \lambda, \tag{2-31}$$

and variance

$$\sigma^2 = \lambda. \tag{2-32}$$

The mean and variance for the probability in the Hospital admission example when calculated by hand would look like,

$$\mu = \lambda = 0.3$$

$$\sigma^2 = \lambda = 0.3$$

Kahoot!

(x1)

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